

**Analyse null keys and null values in collection-based APIs. Develop a policy for an enterprise project and justify where null should be accepted, rejected or replaced with Optional-like semantics. [Analyse/Evaluate]**

### **Aim**

To analyse the handling of null keys and null values in Java collection-based APIs and develop an enterprise policy that specifies where null should be accepted, rejected, or replaced using Optional-like semantics.

- **PSEUDO CODE:**

- Start
- Create an empty Map
- Read Key and Value
- If Key is NULL

Display "Null key rejected"

- Else If Value is NULL

Display "Null value rejected"

- Else

Insert Key and Value into the Map

- End If
- Search for a Key
- If Key exists
- Return Value
- Else

Return Optional.empty (or "Value not found")

- End If
- Display Map
- Stop

**JAVACODE:**

```
import java.util.HashMap;
import java.util.Map;
import java.util.Optional;
public class Main {
    public static void main(String[] args) {
        Map<String, String> map = new HashMap<>();
        // Add entries
        addEntry(map, "E101", "Karthik");
        addEntry(map, null, "Rahul");
        addEntry(map, "E102", null);
        // Display map
        System.out.println("Map: " + map);
        // Search for existing key
        Optional<String> value1 = getValue(map, "E101");
        System.out.println("E101 : " + value1);
        // Search for missing key
        Optional<String> value2 = getValue(map, "E999");
        System.out.println("E999 : " + value2);
    }
    // Method to add entries
    public static void addEntry(Map<String, String> map, String key, String value) {
        if (key == null) {
            System.out.println("Null key rejected.");
        } else if (value == null) {
            System.out.println("Null value rejected.");
        } else {
            map.put(key, value);
            System.out.println("Entry added.");
        }
    }
}
```

```

    }
    // Method to return Optional instead of null
    public static Optional<String> getValue(Map<String, String> map, String key) {
        return Optional.ofNullable(map.get(key));
    }
}

```

## OUTPUT:

```

C:\Users\satya\Desktop\javaprogram\CO2>ja
Entry added.
Null key rejected.
Null value rejected.
Map: {E101=Karthik}
E101 : Optional[Karthik]
E999 : Optional.empty

```